

Develop investigable questions, reasoned predictions and hypotheses to test relationships and develop explanatory models

Learning intention

- develop investigable questions, reasoned predictions and hypotheses to test relationships and develop explanatory models.

Teach and model

- developing investigable questions to explore an explanatory model.
- discussing what is needed for a question to be investigable or a prediction to be reasoned.
- discussing why a scientific hypothesis has to be able to be supported or refuted through evidence.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.